

# Literature Review

## *EuroSAT Land Use Classification: Multispectral (13-Band) Approach vs. Prior RGB-Based Work*

### 1. Existing Approaches (Prior Work)

The majority of published work on the EuroSAT dataset uses only the RGB subset of the data (3 channels), since it is the most convenient format for standard CNN architectures pretrained on natural images (e.g. ImageNet weights). These studies typically load the RGB JPG/PNG version of EuroSAT, resize it to fit a pretrained backbone (ResNet, VGG, EfficientNet), and fine-tune on the 10 land-use classes. Reported accuracies in this line of work commonly range from 95-98% on the held-out RGB test set.

#### Limitations of the RGB-only approach:

- Discards 10 of the 13 Sentinel-2 spectral bands, losing information in the Red Edge, NIR, SWIR, and atmospheric correction bands.
- Cannot compute vegetation/water indices (NDVI, NDWI) that rely on Near-Infrared and Red bands, since NIR is not present in RGB.
- Vegetation and crop-health distinctions (e.g. AnnualCrop vs. PermanentCrop vs. Pasture) rely heavily on NIR reflectance, which RGB models cannot see.
- Performance is often inflated by visual/textural cues rather than true spectral separability, which can fail to generalize to real Sentinel-2 scenes outside the curated RGB crop.

### 2. Our Approach (Multispectral, 13-Band + NDVI)

Rather than the RGB subset, this project uses the full 13-band Sentinel-2 GeoTIFF (.tif) version of EuroSAT for every image. All 13 spectral bands (Coastal Aerosol, Blue, Green, Red, Red Edge 1-4, NIR, Water Vapor, and the SWIR/Cirrus bands) are read directly from the .tif files and resized to 64x64 while preserving all channels.

During preprocessing, the Normalized Difference Vegetation Index (NDVI) is computed from the Red band (band 4) and the NIR band (band 8) as:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red} + \epsilon)$$

This NDVI layer is appended as a 14th channel, so every model in this project (SimpleCNN, ResNet50, EfficientNet, SpectrumNet) is trained on a 64x64x14 input rather than 64x64x3. Geometric data augmentation (rotation, zoom) is applied only to the training split, and is spatial-only so it applies cleanly across all 14 channels.

### 3. Comparative Analysis

| Aspect            | Prior Work (RGB)           | This Project (13-Band + NDVI)                     |
|-------------------|----------------------------|---------------------------------------------------|
| Input channels    | 3 (Red, Green, Blue)       | 14 (13 Sentinel-2 bands + NDVI)                   |
| Source format     | RGB JPG/PNG crop           | Native Sentinel-2 GeoTIFF (.tif)                  |
| Spectral coverage | Visible spectrum only      | Visible + Red Edge + NIR + SWIR/atmospheric bands |
| Vegetation index  | Not possible (no NIR band) | NDVI computed and used as an extra channel        |

| Aspect                    | Prior Work (RGB)                        | This Project (13-Band + NDVI)                                                      |
|---------------------------|-----------------------------------------|------------------------------------------------------------------------------------|
| Pretrained backbones      | Directly usable (3-ch ImageNet weights) | Backbones adapted to 14-ch input (no ImageNet init)                                |
| Class separability        | Relies on color/texture only            | Adds physical reflectance signal (e.g. crop vs. pasture health)                    |
| Real-world inference risk | Lower band-mismatch risk                | Higher risk from normalization/band-order mismatch; needs consistent preprocessing |

## 4. Summary

Compared to prior RGB-only EuroSAT studies, this project retains the full spectral resolution of Sentinel-2 imagery and augments it with a derived NDVI channel, giving the models access to vegetation and moisture signals that are invisible in RGB. This comes at the cost of losing the ability to reuse ImageNet-pretrained weights directly, and it raises the practical requirement that any inference pipeline reproduce the exact same band order, NDVI formula, and normalization used during training.